

Bonmoo, Koo

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CAREER OBJECTIVE

Motivated researcher aiming to contribute to energy-aware resource management and efficient database systems through innovative software design

EDUCATION

KAIST (2024.3~) Daejun, S.Korea

Integrated M.S./Ph.D. Program in Electrical Engineering

Konkuk University (2018,2021,2022,2023) Seoul, S.Korea

B.S in Computer science Engineering (GPA : 4.18/4.5) Mar.2018-Feb.2023

Scholarship : Konkuk University Scholarship for academic excellence (2018,2021,2022,2023)

Course Taken: System Programing, OS, Database System, Software Architecture, Computer Architecture, Algorithm,

WORK EXPERIENCE

Naver Cloud Feb.2026~Present

- High-performance distributed file system development

KAIST OS Lab Mar.2024~Present

- Graduate Researcher

Konkuk University Data Centric Computing Laboratory Jun.2022-Dec.2023

- Undergraduate Researcher

- Summarized various paper in data system association and submitted as government research achievement

- Implemented range index structure with RDMA (Remote Direct Memory Access) for disaggregated memory

Korean Army Jun.2019 ~ Dec.2020

-Korea Military Academy Cadet training assistant

PROJECT

Designing Hashtable For Range Query

Mar.2023~Dec.2023

- Graduation Project
- Build In-memory Hashtable which supports range query

PUBLICATIONS

"Directory-Granularity Filesystem Journaling for CMM-H SSDs", FAST, Feb 2025 (3rd Author)

"ESL: A High-Performance Skiplist with Express Lane", Applied-Science , Sep 2023 (2nd Author)

- Designed Cacheline-friendly skiplist using Express Lane for search layer

"Converting Concurrent Range Index Structure to Range Index Structure for Disaggregated Memory", Applied-Science, Sep 2023 (1st Author)

- Studied characteristic for disaggregated memory
- Suggested SPREAD protocol and implemented current BLinkTree into range index structure for disaggregated memory using one-sided RDMA verbs

SKILLS

Coding : C/C++ (5/5), Python (4/5)

- Lots of experience building data structure and using one-sided verb RDMA communication between multiple devices
- Experience in kernel coding

Hobbies : Skiing , Tennis